

Alberto Angeloni

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Education

Università degli Studi di Padova September 2026
Master degree in Computer Science Padova, Italy

- **Relevant Coursework:** Advanced Algorithms, Software Verification (Static Analysis and Abstract Interpretation), Formal Methods for Cyber-Physical Systems, Real-Time Kernels and Systems, Functional Languages, Runtimes for Concurrency and Distribution

Università degli Studi di Padova December 2023
Bachelor degree in Computer Science (93/110) Padova, Italy

- **Relevant Coursework:** Algorithms and Data Structures, Probability and Statistics, Software Engineering, Operating Systems, Database Systems, Linear Algebra and Discrete Mathematics

I.I.S J.F. Kennedy July 2019
High School Degree (72/100) Monselice, Italy

- **Relevant Coursework:** Programming and Databases, Systems and Networks, Web Technologies, Information Systems and Telecommunications Design, Project Management, Mathematics

Experience

Almaviva Mar 2026 – Sep 2026
Software Analyst Intern Padova, Italy

- Contributed to the SI.Ter regional healthcare programme through requirements analysis, interoperability flowcharts, functional proofs of concept, and technical assessments for healthcare organisations in Veneto.
- Designed and implemented a Python framework that generates synthetic FHIR R4 patients and executes multi-step RVE and IHE workflows over REST and SOAP, with SAML/JWT propagation, mTLS, and asynchronous traffic generation.
- Built a measurement pipeline for correlated JSONL logs and latency, throughput, reliability, and concurrency metrics; validated it on 74 complete live runs covering 158,449 flows and 299,534 responses.

CWBI July 2023 – Sep 2023
Software Developer Intern Padova, Italy

- Developed a Dart/Flutter module to register graphometric-signature tablets with an external service and manage the full Namirial SDK licence lifecycle, from activation and validity checks to expiry and renewal.
- Connected Flutter to the native Android/Kotlin signing SDK through platform channels, handling licence activation, mandatory signature fields, partial signing, and PDF signing outcomes.
- Integrated the module into xMed, a tablet application that lets private clinics retrieve informed-consent documents, collect biometric graphometric signatures, and return signed PDFs to a REST back office.

Projects

Software Verification Analyzer | *Rust, Abstract Interpretation*

- Built a lexer, parser, and evaluator for a small imperative language, following the abstract-interpretation work of Cousot and Miné; implemented denotational semantics and interval analysis with widening and narrowing.

Functional Language Compiler | *Lambda Calculus, Static Typing*

- Developed a compiler for a minimal functional language based on the lambda calculus, supporting let bindings, lambda abstractions, function application, static type checking, and type inference.

Raft Consensus for Seismic Sensor Networks | *Ada, Distributed Systems*

- Implemented Raft for a simulated seismic sensor network, covering leader election, message passing, log replication, and node failures to preserve data availability and consistency under faults.

Ravenscar-to-RTIC Real-Time Port | *Ada, Rust, RTIC*

- Ported an Ada Ravenscar application to embedded Rust on RTIC, mapping periodic, sporadic, and aperiodic workloads to priority-based tasks and comparing the timing performance of the two implementations.

Formal Verification and Controller Synthesis | *Python, PyNuSMV, CIF*

- Implemented a PyNuSMV model checker for reactivity formulas, using symbolic reachability and cycle detection to verify properties and construct counterexamples.
- Modelled a three-floor elevator with Extended Finite Automata in CIF and synthesized a supervisor that enforces its movement and service requirements.

Technical Skills

Languages: C++, Python, Rust, Ada, Dart, Kotlin, SQL, F#, TypeScript, HTML, CSS, JavaScript

C++: Standard Library (STL), RAII, Smart Pointers, Object Lifetime, Memory Management, Move Semantics, Templates, Polymorphism, Virtual Dispatch

Frameworks & Tools: Flutter, Android SDK, RTIC, PyNuSMV, Eclipse ESCET/CIF, Git, $\LaTeX$, SoapUI

Interoperability & Security: FHIR R4, IHE profiles, REST, SOAP, SAML, JWT, mTLS

Computer Science: Static Analysis, Abstract Interpretation, Compiler Construction, Type Systems, Distributed Consensus, Real-Time Systems, Model Checking, Supervisory Control